

KANDAHAR AP, Afghanistan

WMO: 409900

Lat: 31.5021N

Lon: 65.8524E

Elev: 1007

StdP: 89.80

Time Zone: 4.50 (AFG)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	-1.0	-19.0	0.8	6.8	-17.0	0.9	7.0	11.9	6.5	11.0	5.3	2.5	60	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	42.2	17.3	41.2	17.1	40.2	16.8	20.6	34.4	18.9	36.8	17.7	38.4	4.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	12.0	28.9	12.9	10.5	21.9	11.1	9.3	19.5	64.2	34.8	57.9	36.5	53.8	38.5	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.3	7.1	-5.3	44.4	2.1	0.7	-6.8	45.0	-8.0	45.4	-9.2	45.8	-10.7	46.4		
(5)			-7.9	21.4	1.7	1.8	-9.1	22.7	-10.1	23.7	-11.1	24.7	-12.3	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	8.1	10.5	16.5	22.2	27.5	31.3	33.3	31.3	26.1	20.5	14.2	8.7		
	DBStd	9.36	3.61	4.38	3.99	3.61	2.81	2.41	1.72	2.18	2.82	3.25	3.63	3.42		
	HDD10.0	197	82	42	2	0	0	0	0	0	0	0	8	63		
	HDD18.3	1075	318	221	83	10	0	0	0	0	1	16	129	297		
	CDD10.0	4175	22	56	203	366	542	638	722	660	483	324	134	24		
	CDD18.3	2012	0	1	27	126	283	388	463	402	234	82	6	0		
	CDH23.3	30874	0	14	334	1573	3971	6266	7774	6324	3341	1153	124	0		
(13)	CDH26.7	19384	0	1	84	714	2344	4216	5410	4181	1930	483	21	0		
(14)	Wind	WSAvg	2.9	3.3	3.9	3.2	3.2	3.4	2.9	2.8	2.6	2.2	2.2	2.3		
(15) Precipitation	PrecAvg	183	31	42	35	20	7	3	6	4	4	4	11	16		
	PrecMax	307	71	93	89	81	42	40	43	25	33	33	62	68		
	PrecMin	72	3	5	1	0	0	0	0	0	0	0	2	0		
	PrecStd	57	20	24	21	18	8	7	11	7	8	8	13	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.8	25.1	31.2	36.2	40.3	43.8	43.3	42.2	38.8	35.2	29.1	21.9		
		MCWB	9.3	11.7	13.4	15.5	16.2	17.4	17.8	17.4	15.9	13.9	12.4	9.3		
	2%	DB	19.3	22.9	29.0	34.3	39.1	42.1	42.4	41.1	37.2	33.9	26.8	19.9		
		MCWB	8.2	10.5	12.9	14.7	16.1	17.0	17.4	17.3	15.2	13.6	11.6	8.2		
	5%	DB	17.2	20.6	27.0	32.9	37.9	41.0	41.8	40.0	36.2	31.8	24.4	18.0		
		MCWB	7.2	9.9	12.0	14.2	15.8	16.8	17.2	16.9	14.8	13.0	10.6	7.4		
	10%	DB	15.1	18.0	24.9	31.0	36.0	39.8	40.8	38.9	35.1	29.2	22.2	16.0		
		MCWB	6.3	9.2	11.3	13.5	15.2	16.3	17.1	16.6	14.4	12.0	9.7	6.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	13.4	16.2	16.5	17.2	23.1	20.7	21.8	20.0	15.0	14.7	11.0		
		MCDB	16.1	19.3	23.6	29.6	35.8	27.6	37.3	34.8	30.7	29.7	20.0	18.0		
	2%	WB	9.6	12.4	14.7	15.7	16.7	19.3	19.2	21.0	18.2	14.1	13.3	9.6		
		MCDB	16.0	18.5	23.1	28.5	35.6	37.0	38.5	33.8	32.2	30.1	19.7	15.6		
	5%	WB	8.6	11.3	13.6	15.2	16.2	17.7	18.3	19.1	15.5	13.3	12.2	8.6		
		MCDB	14.9	17.7	22.2	27.7	34.8	39.3	39.1	36.1	35.7	29.8	20.6	15.0		
	10%	WB	7.6	10.3	12.7	14.6	15.7	16.8	17.6	17.2	14.6	12.5	11.2	7.7		
		MCDB	13.4	16.6	21.2	26.9	33.6	38.7	39.2	37.9	34.6	28.8	20.3	14.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	10.6	13.0	14.0	14.8	15.5	14.7	14.8	17.2	16.5	14.2	13.1		
		MCDBR	14.5	13.4	15.2	16.7	17.2	16.8	15.6	15.3	18.4	18.5	16.5	15.0		
		MCWBR	7.5	5.4	5.0	5.6	6.4	6.5	6.2	6.3	7.7	8.2	7.9	7.6		
		MCDBR	11.3	10.7	12.3	13.5	15.1	14.8	14.2	11.5	16.1	16.8	12.6	11.8		
	5% WB	MCWBR	6.5	5.1	4.6	4.6	5.4	5.8	6.3	5.6	7.0	7.4	6.3	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.392	0.437	0.477	0.468	0.459	0.524	0.488	0.388	0.358	0.340	0.317		
	taud		2.264	2.073	1.941	1.870	1.899	1.922	1.777	1.855	2.090	2.169	2.245	2.329		
	Ebn at Noon		880	852	839	821	830	833	778	801	876	877	863	881		
	Edh at Noon		110	147	181	202	198	193	222	202	153	131	111	98		
(44) All-Sky Solar Radiation	RadAvg		3.56	4.24	5.63	6.82	7.84	8.39	7.80	7.33	6.66	5.50	4.14	3.49		
	RadStd		0.26	0.39	0.34	0.28	0.24	0.24	0.33	0.33	0.16	0.14	0.20	0.24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page